



Lab 05: Routing Between Different Networks

Router0 (2911) · Switch0 · Switch1 · PC0 · Laptop0 · PC1

Lab #05

Topic: Inter-Network Routing

Tool: Cisco Packet Tracer

Cert: CompTIA Network+

1. Lab Objectives

- Connect two 2960-24TT switches to Router0 (2911) on separate Gigabit interfaces.
- Place PC0 and Laptop0 on Switch0 (subnet 192.168.1.0/24) and PC1 on Switch1 (192.168.2.0/24).
- Router0 provides Layer 3 routing between the two switch networks.
- Verify cross-subnet ping from PC0 and Laptop0 to PC1.
- Relate to Network+ objective 1.4: Routing concepts and path determination.

2. Network Topology

Router0 (2911) sits at the top. Gig0/0 connects to Switch0 (2960-24TT) serving PC0 (Fa0/1) and Laptop0 (Fa0/3). Gig0/1 connects to Switch1 (2960-24TT) serving PC1 (Fa0/1). Two /24 subnets are used — one per switch.

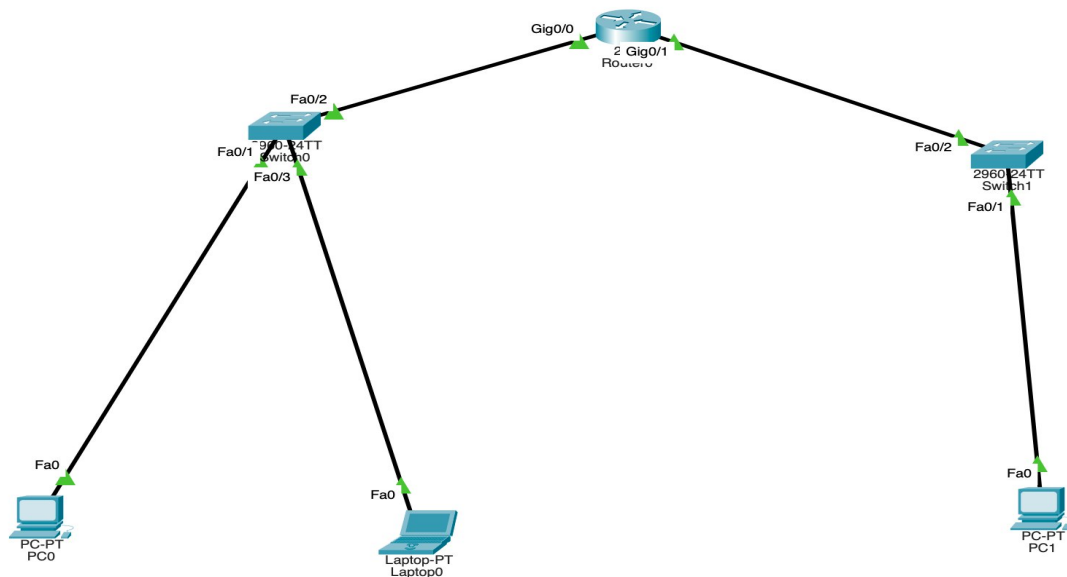


Figure 1 – Router0 connecting Switch0 (PC0, Laptop0) and Switch1 (PC1)

3. IP Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway	Network
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Router0	Gig0/0	192.168.1.1	255.255.255.0	—	192.168.1.0/24
Router0	Gig0/1	192.168.2.1	255.255.255.0	—	192.168.2.0/24
PC0	Fa0	192.168.1.10	255.255.255.0	192.168.1.1	192.168.1.0/24
Laptop0	Fa0	192.168.1.20	255.255.255.0	192.168.1.1	192.168.1.0/24
PC1	Fa0	192.168.2.10	255.255.255.0	192.168.2.1	192.168.2.0/24

4. Step-by-Step Configuration

4.1 Router0 – Both Interfaces

```
Router0(config)#interface GigabitEthernet0/0
Router0(config-if)#ip address 192.168.1.1 255.255.255.0
Router0(config-if)#no shutdown
Router0(config-if)#exit

Router0(config)#interface GigabitEthernet0/1
Router0(config-if)#ip address 192.168.2.1 255.255.255.0
Router0(config-if)#no shutdown
Router0(config-if)#exit

Router0#write memory
```

4.2 PC and Laptop Static IPs

PC0 – IP / Mask / GW	192.168.1.10 / 255.255.255.0 / 192.168.1.1
Laptop0 – IP / Mask / GW	192.168.1.20 / 255.255.255.0 / 192.168.1.1
PC1 – IP / Mask / GW	192.168.2.10 / 255.255.255.0 / 192.168.2.1

5. Verification

```
PC0>ping 192.168.1.20      ! PC0 to Laptop0 – same subnet (via switch)
PC0>ping 192.168.2.10      ! PC0 to PC1 – cross-subnet (via Router0)
Laptop0>ping 192.168.2.10  ! Laptop0 to PC1 – cross-subnet (via Router0)
Router0#show ip route      ! C 192.168.1.0 and C 192.168.2.0 must appear
```

6. Key Takeaways

- One router with two interfaces = two subnets and two separate broadcast domains.
- PC0 and Laptop0 share 192.168.1.0/24 — the switch handles their traffic without the router.
- PC1 is on 192.168.2.0/24 — cross-subnet traffic must pass through Router0.
- Connected routes are added to the routing table automatically when interfaces are up/up.



- Always configure the correct gateway on each PC — wrong gateway = no routing.